Agricultural Production and Harvest Tracking System

Real Rwanda Crop Benchmark Analytics

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Problem Statement

Rwanda already has agricultural digital services, including the *Smart Nkunganire System*, powered by BK Techouse (BK Techouse, n.d.). A study of 110 maize farmers in Gatsibo and Nyagatare examined post-harvest losses and input supply (Gaspard et al., 2021). District planners and cooperative advisers need consistent, quality-checked crop benchmarks to compare results across seasons and areas. Without them, missing values and incompatible units can be mistaken for real performance differences, misdirecting planning and advice. This project addresses that complementary analytics gap; it does not claim that Rwanda lacks agricultural systems.

Proposed Solution

Build an Oracle database and Oracle APEX analytics application centered on the Rwanda HarvestStat Africa extract. Users will compare aggregate crop-area, production, and yield observations by crop, historical area, season, and available year, with completeness and unit-quality warnings. The registered title is retained, but this phase is specifically benchmark analytics - not individual farm or harvest tracking.

The extract has 13,725 rows, 9,123 populated values, and 4,602 missing values (33.5%). It has no farmer, cooperative, farm, or plot identifiers. No observation will be attached to a fictitious farm; operational farm-level tables remain out of scope without an authorized, consented source.

Data Source and Fit

HarvestStat Africa is a harmonized subnational crop-statistics resource with the crop, season, area, indicator, value, and unit fields needed for benchmark comparisons (HarvestStat Consortium, n.d.; Lee et al., 2025). The Rwanda extract has 38 crop/product names, 42 geographic-unit labels, indicators for area planted, quantity produced, and yield, and mixed units (ha, t, MT/ha) that require validation. HDX identifies HarvestStat Consortium as the dataset organization; dataset metadata lists the license as Other; an earlier resource listing identified CC0 1.0, but the current resource API has no file-level license value. Preserve attribution and verify exact file terms before redistribution.

Only eight years occur: 1984, 2006, 2008, 2009, 2010, 2012, 2013, and 2017. There is no 1990s coverage and only one year in the 1980s. This is sparse historical coverage, not a continuous time series. The dashboard will focus on crop, historical-area, and season comparisons, not long-term yield trends.

Planned Tables

SOURCE_DATASET and IMPORT_BATCH preserve provenance; ADMIN_AREA, CROP, SEASON, and INDICATOR are dimensions; AGG_CROP_STATISTIC stores sourced aggregate facts; DATA_QUALITY_ISSUE records missing, invalid, duplicate, or rejected observations; DATA_CORRECTION_REQUEST routes proposed corrections; AUDIT_EVENT records approved changes. Farm, plot, planting, and harvest entities are outside this phase.

Planned PL/SQL Features

- Package PKG_HARVEST_BENCHMARK centralizes import, quality, correction, and APEX report logic.
- Procedure P_VALIDATE_STAGE validates source keys, period/geography, indicator/unit pairs, and duplicates before batch acceptance; rejected rows are logged.
- Procedure P_REFRESH_QUALITY_SUMMARY uses cursor C_QUALITY_GAPS to refresh crop/indicator/decade completeness metrics for the APEX quality report.
- Procedure P_APPLY_APPROVED_CORRECTION verifies reviewer authorization and request status, then applies only approved changes through protected application context.
- Function F_COMPLETENESS_PCT returns nonmissing-value percentage for the selected filters.

Function F_SAFE_YIELD calculates quantity / area only for compatible units and positive area; otherwise it raises a custom exception.

Rule trigger TRG_CORR_REQUEST_VALIDATE checks correction target, rationale, proposed value, and indicator/unit consistency.

Restriction trigger TRG_AGG_RESTRICT_DIRECT_DML blocks direct edits/deletes to curated benchmark rows; an authorized package operation is required for an approved correction.

Audit trigger TRG_AGG_AUDIT records old/new values, user, timestamp, and correction-request ID.

Custom exceptions E_INVALID_UNIT, E_CORRECTION_NOT_FOUND, E_CORRECTION_NOT_APPROVED, and E_UNAUTHORIZED_REVIEWER explain rule, request, and permission failures.

Innovation

Option: Analytics dashboard. Tool: Oracle APEX connected to Oracle. Users filter real aggregate observations by crop, historical area, season, and available year, and inspect missingness and unit warnings. PL/SQL supplies validated imports, completeness summaries, and unit-safe calculations; the demonstration uses imported data, not a static mock-up.

References

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